Salchi Metalcoat S.r.l.
Coatings and inks for coil and metal packaging
Viale dell'Industria 3/A - 3/B
20875 Burago di Molgora (MB) - Italy
T + 39 039 6251824 / F + 39 039 6251829
info@salchirh.it / salchimetalcoat.com



BEST TECHNOLOGY SOLUTION with schekolin
Hochwertige Lacke und Farben
High quality lacquers and paints

*Rev. 01 – 10.06.2022*

# 422 0 903 — 5761000001 SCHEKOSOL WH BC

**Suggested use :** Toned White Basecoat for external of Monobloc aerosol cans
**Type :** Polyester, BPA NI
**Solvents :** Aromatic Hydrocarbons, Alcohols, Glycolethers, Glycolesters

## PHYSICAL CHARACTERISTICS

| | | | |
|---|---|---|---|
| **Delivery Viscosity :** | (DIN 6 cup at 20 °C) | 75 ± 5 | s. |
| **Application Viscosity :** | (DIN 6 cup at 20 °C) | 70 – 100 | s. |
| **Density :** | (Pycnometer at 20 °C) | 1420 – 1460 | g/L |
| **Solid contents :** | (30 min. at 180 °C) | 71 ± 2 | % |
| **Shelf life :** | (At about 25 °C) | 8 months | |
| | *-store in cool and dry place in original packaging-* | | |

## RECOMMENDED APPLICATION PROCEDURE

| | | |
|---|---|---|
| **Apply by :** | Roller coater | |
| **Apply over :** | Monobloc Aerosol cans | |
| **Preliminary treatment :** | Usual pre-treatment | |
| **Dilution :** | Usually not needed, 1 – 2 % if necessary | |
| **Reducer :** | 506 9 003 | |
| **Wet film weight :** | 32 – 43 $g/m^2$ | |
| **Dry film weight :** | 12 – 16 µm | |
| **Curing conditions :** | Pre-drying | 3 – 7 min. at 100 – 120 °C |
| | Intermediate drying | 5 – 7 min. at 150 – 160 °C |
| | Final-drying | 6 – 10 min. at 160 – 180 °C |

## REMARKS :

- Very good elasticity and good levelling
- High spreading and body
- Good sweating effect
- Resistant to steam bath
- Excellent mechanical properties
- Suitable for monobloc aerosol cans up to diameter 66 mm
- Recommended procedure: 1st Schekosol BC 4220903, 2nd Printing, 3rd Suitable overprint varnish, 4th Final drying

**DEVELOPMENT AND QUALITY CONTROL TEST ARE PERFORMED ON THE FOLLOWING SUBSTRATES:**
**ETP E 2.0/2.0 or E 2.8/2.8 upon availability, stone finished, 310 or 311 passivated; ECCS with min. 50 mg/sqm of Chrome; ALUMINIUM 8011 alloy, H14 temper**

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**Rev. 01 – 30.10.2023**

# 400 9 901 57802 SCHEKOSOL INT PROT

| | |
|---|---|
| **Suggested use :** | Gold internal protective lacquer for Aerosol cans made of aluminium |
| **Type :** | Epoxy resins combination |
| **Solvents :** | Aromatic Hydrocarbons, Alcohols, Glycolethers, Glycolesters |

## PHYSICAL CHARACTERISTICS

| | | | |
|---|---|---|---|
| **Delivery Viscosity :** | (Din 4 cup at 20 °C) | 85 ± 5 | s. |
| **Application viscosity:** | (Din 4 cup at 20 °C) | 85 ± 5 | s. |
| **Density :** | (Pycnometer 20 °C) | 950 – 990 | g/l |
| **Solid Contents :** | (30 min. at 200 °C) | 36 ± 2 | % |
| **Shelf life :** | (At about 20 °C) | 9 months | |

*- store in a cool and dry place in original packaging-*

## RECOMMENDED APPLICATION PROCEDURE

| | |
|---|---|
| **Apply by :** | Spraying |
| **Substrate :** | Pre-treated Aerosol cans made of aluminium |
| **Dilution :** | If needed |
| **Reducer :** | 506 9 901 |
| **Wet film weight :** | 37,0 – 44,0 g/m$^2$ |
| **Dry film weight :** | 10,0 – 12,0 µm |
| **Curing conditions :** | 5 – 10 minutes at 220 - 240°C |

## REMARKS :

- Universally applicable
- Higher security by tapering cans with large diameters.
- Recommended procedure: 1 or 2x of gold protective lacquer 4009901

**DEVELOPMENT AND QUALITY CONTROL TEST ARE PERFORMED ON THE FOLLOWING SUBSTRATES:**
**ETP E 2.0/2.0 or E 2.8/2.8 upon availability, stone finished, 310 or 311 passivated; ECCS with min. 50 mg/sqm of Chrome; ALUMINIUM 8011 alloy, H14 temper**

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High quality lacquers and paints



**Rev. 00 – 22.05.2018**

# 400 4 902 44850 SCHEKOSOL INT BEIGE

| | |
|---|---|
| **Suggested use :** | Beige Lacquer for interior protection of Aluminium aerosol cans |
| **Type :** | Epoxy resin combination |
| **Solvents :** | Aromatic Hydrocarbons, Alcohols, Glycolethers, Glycolesters |

## PHYSICAL CHARACTERISTICS

| | | | |
|---|---|---|---|
| **Delivery Viscosity :** | (DIN 4 cup at 20 °C) | 60 ± 5 | s. |
| **Application Viscosity :** | (DIN 4 cup at 20 °C) | 60 ± 5 | s. |
| **Density :** | (Pycnometer at 20 °C) | 1020 – 1050 | g/l |
| **Solid contents :** | (30 min. at 200 °C) | 38 ± 2 | % |
| **Shelf life :** | (At about 25 °C) | 8 months | |
| | *-store in cool and dry place in original packaging-* | | |

## RECOMMENDED APPLICATION PROCEDURE

| | |
|---|---|
| **Apply by :** | Spraying |
| **Apply over :** | Aerosol Cans |
| **Preliminary treatment :** | Usual pre-treatment |
| **Dilution :** | Usually not needed, 1 – 2 % if necessary |
| **Reducer :** | 506 9 901 |
| **Wet film weight :** | 41 – 48 g/m$^2$ |
| **Dry film weight :** | 10 – 12 g/m$^2$ |
| **Curing conditions :** | 3' at 280 °C to 10' at 240 °C |

## REMARKS :

- Universally applicable
- Higher security by tapering cans with large diameters
- Recommended procedure: 1x or 2x SCHEKOSOL INT BEIGE lacquer 4004902

**DEVELOPMENT AND QUALITY CONTROL TEST ARE PERFORMED ON THE FOLLOWING SUBSTRATES:**
**ETP E 2.0/2.0 or E 2.8/2.8 upon availability, stone finished, 310 or 311 passivated; ECCS with min. 50 mg/sqm of Chrome; ALUMINIUM 8011 alloy, H14 temper**

# SunAltec™ MB PLUS Series

## 1. Description

**The SunAltec MB PLUS** Series for Monobloc Cans is a versatile, mineral oil free, Alkyd based Ink System which has been modified for optimum Performance not only on the new Generation of Production Lines but also for the today's Basecoat Technology.

## 2. Product features

- The inks are supplied in a ready to Print form
- Mineral oil free
- The Base Ink System is ideal for mixing any Spot Color directly on site.
- Highly concentrated Inks allow high quality printing with low Film thickness
- The Color Matching Database is readily available for x-rite IF Systems either via TurnKey or Local

## 3. Product Suitability

### 3.1 Applications

**The SunAltec MB PLUS** inks are intended for use in the following areas:

- Aluminium Monobloc Cans
- Rigid Aluminium Tubes
- Screw Cap Cans
- Shakers
- ROPP (Skirt Printing)
- Industrial Products i.e. Cartouches

For any other end uses not listed, please refer to your local Sun Chemical representative for specific details and advice.

Please note, **the Inks are intended for Outside Printing only**

**The SunAltec MB PLUS** inks are not suitable for following Applications

- Food packaging without functional Barrier
- Metallic inks should not be used for pasteurisation or sterilisation applications
- Carefully select the inks based on their Fastness Properties for Products that are sterilised.

A Test on the finished product using the real Process is strongly recommended every time as all Parameters of the Packaging are playing a role

### 3.2 Substrate

**SunAltec MB PLUS** inks are suitable for the following substrates and structures-:

- Aluminium
- Tin

## 4. Colour Range

**SunAltec MB PLUS** inks are available as bespoke finished inks.

| Color code | Designation | Container: vacuum cans of | Light | Alcohol | Solvent | Alkali | Temperature |
|---|---|---|---|---|---|---|---|
| 169167 | TRANSPARENT WHITE | 2.5 Kg | * | + | + | + | 180 °C |
| 169766 | OPAQUE WHITE | 2.5 Kg | * | + | + | + | 160 °C |
| 168390 | YELLOW G/S | 2.5 Kg | 5 | + | + | + | 200 °C |
| 168391 | YELLOW | 2.5 Kg | 5 | + | + | + | 200 °C |
| 168392 | YELLOW R/S | 2.5 Kg | 5 | + | + | + | 200 °C |
| 168393 | ORANGE | 2.5 Kg | 5 | + | + | + | 160 °C |
| 168394 | WARMRED | 2.5 Kg | 7 | + | + | + | 200 °C |
| 168395 | RUBINE RED | 2.5 Kg | 5 | + | + | + | 180 °C |
| 168396 | MAGENTA | 2.5 Kg | 5 | + | + | + | 180 °C |
| 168397 | RHODAMINE RED | 2.5 Kg | 7 | + | + | + | 180 °C |
| 168398 | PURPLE | 2.5 Kg | 6 | + | + | + | 180 °C |
| 168399 | VIOLET | 2.5 Kg | 7 | + | + | + | 200 °C |
| 168400 | BLUE | 2.5 Kg | 7 | + | + | + | 200 °C |
| 168401 | BLUE | 2.5 Kg | 8 | + | + | + | 200 °C |
| 168402 | BLUE (REFLEX) | 2.5 Kg | 7 | + | + | + | 200 °C |
| 168403 | GREEN | 2.5 Kg | 8 | + | + | + | 200 °C |
| 168404 | BLACK | 2.5 Kg | 8 | - | - | + | 200 °C |
| 168404/1 | BLACK Conc. | 2.5 Kg | 8 | - | - | + | 200 °C |

**Fastness to light** — **1** (very poor) – **8** (best)
**Fastness to solvents, alcohol and alkali** — **+** suitable, **+/-** to try out, **-** not suitable

*Values calculated according to pigment suppliers information. Values refer to a solid printing without overprint varnish. These values may decrease when colour strength is reduced, when colours are intermixes, when printing half-tones and/or when an overprint varnish is applied*

## 5. General Handling

### 5.1 Storage

**SunAltec MB PLUS** inks are not flammable.

Shelf life: **36 months*** from the Day of Production

**For unopened Tins, stored in dry conditions at 15-30 °C*

*Please note,*
*From practical experiences we know that inks could be printed without any problems, even after a storage time of more than 5 years. Since the storage conditions are however of crucial importance, we cannot give you a general guaranty for such a long time. Viscosities tend to increase over time, if so, it can be adjusted with 166777*

### 5.2 Waste disposal

Care should be exercised in the disposal of printing ink waste. This should be carried out in accordance with good industrial practice, observing all the appropriate regulations and guidelines. For more specific handling advice refer to the safety data sheet (SDS).

## 6. Printing Conditions

### 6.1 Printing

To be applied onto previously coated Substrate by dry offset printing. Suitable for polyester- and polyurethane enamels.

**SunAltec MB PLUS** inks are curing at elevated Temperatures. Complete drying is achieved at temperatures of

**150–180 $^0$C for 6 to 10 Minutes**

Properly cured inks should resist 20 double rubs with Acetone or MEK.

### 6.2 Auxiliaries

For insufficient drying, additional dryer can be added.

The ink is supplied at printing viscosity, has good press stability, transfer and image reproduction. However, if necessary the ink can be adjusted to the requirements of job and machine by adding suitable and recommended additives.

## 7. End-use safety

All Sun Chemical products are formulated to the latest version of the CEPE/EuPIA exclusion list. This excludes the use of substances classified as **carcinogenic, mutagenic and toxic for reproduction (CMR 1 and 2)** or labelled (T) according to the Dangerous Substances Directive 67/548/EEC, substances classified as very toxic (T+) or toxic (T) and pigments based on compounds of Sb, As, Cd, Cr (VI), Pb, Hg, Se. The use of certain dyes, solvents, plasticisers and miscellaneous materials are also excluded. A copy of the document is available on request or directly from the EuPIA website, www.eupia.org

All Sun Chemical products are designed with end-use safety built-in. CoMB Plusination of Sun Chemical products with products from other manufacturers may affect the suitability of the Sun Chemical product for specific end uses and/or applications. It is advised that in such circumstances the user undertakes a detailed risk assessment to ensure that safety is not compromised

**Information on substances of very high concern**
Sun Chemical's products intended for the EU market do not contain any known amounts of substances of very high concern (SVHCs) at more than 0.1%*, which could be subject to potential future authorisation process (as outlined in REACH Art 56). For customers outside of the EU, please request our current EU label and SDS when considering exports to the EU based upon our non-EU products. *or the lowest concentration limit specified in Part 3 of Annex VI to Regulation (EC) No 1272/2008 which results in the classification as a SVHC

**REACH - Identified uses and Use Descriptor System**

Sun Chemical will request suppliers to include or pass up the supply chain the following uses for inks related materials:

| | Product Category | Sector of Use | Process Category | Article Category | Environmental Release Category |
|---|---|---|---|---|---|
| Manufacture | PC 18 | SU 10 | PROC 1, 2, 3, 5, 8a, 8b, 9 | N/A | ERC 2 |
| Application | PC 18 | SU 3 | PROC 2, 3, 4, 5, 8b, 10 | N/A | ERC 4, 5 |

Ref: ECHA Guidance on information requirements and chemical safety assessment, version 1.2; May 2008

## 8. Disclaimers

The **SunAltec MB PLUS** ink range is only suitable for use on the non-food-contact side of food packaging provided that they are applied under the relevant Good Manufacturing Practices (GMP) and according to the recommendations of this Technical Data Sheet.

The printer, converter and the packer or filler entity have the legal responsibility to ensure that the finished article is fit for the intended purpose (s) and that the ink and coating components do not migrate** into the food at levels that exceed legal and industry requirements as outlined in the EU Framework Regulation (EC) No 1935/2004 and the GMP Regulation (EC)No 2023/2006 and for film applications the Plastics Directive 2002/72/EC .We recommend that the finished packaging is tested under appropriate representative conditions of use if there are any doubts regarding compliance.

** In the specific situation of **TUB**, **PES** and **SUNALTEC MB PLUS** ink series, these are metal deco inks intended to be applied to the exterior of pre-formed aluminium containers with a wall thickness greater than 12 µm. This constitutes a barrier to migration, and the shape of the container does not allow for any set-off or transfer from the print to the foodstuff. Consequently, according to Article 26*e* paragraph 2, the provisions of the Ordinance (listing in Annex 1 or 6 and observing the relevant migration limits) do not apply to the use of these inks under these packaging scenarios.

## 9. Technical Assistance / Contacts

For further information, please contact your local Sun Chemical team under switzerland@sunchemical.com

**Sun Chemical AG**
Grindlenstrasse 3
CH-8954 Geroldswil
Switzerland
Tel. +41 44 749 50 50
switzerland@sunchemical.com

SAPILUB

# LUBRIMET GR8

Slug lubricant

# TECHNICAL DATA SHEET

## Product Information

**LUBRIMET GR8** is a further development of slug lubricant manufactured by SAPILUB. This new lubricant has an improved extrusion performance what leads into less scratches and extension of tool life. **LUBRIMET GR8** is based on natural, renewable raw materials, is free of heavy metals and biologically degradable.

## Advantages

- **Wall ironing process:** Eliminates or reduce the roll over issue. Better strip off performance of aluminum cans.
- **Straight extrusion:** Dust free and even distribution of the lubricant on the surface of the slugs.

## Product Characteristics

| | |
|---|---|
| **Appearance:** | white |
| **Physical aspect:** | soft, greasy |
| **Characteristics:** | Solid content: 51.5 - 55.5 % |
| | VOC content: 44.5 - 48.5 % |
| **Composition:** | saturated fatty acid salts |

## Application

85 - 130 gram **LUBRIMET GR8** for 100 kilogram aluminum slugs (please check recommendation).

## Tumbling

15 - 20 minutes (ideal at approx. 20 rpm). The tumbling procedure has to be adapted according to the type of slug tumbler. After tumbling, a rest time of approx. 5 minutes is recommended before feeding the press.

## Tumbling procedure

1. Put half a load of the slugs into the tumbler
2. Spread **LUBRIMET GR8** throughout the load
3. Add the remaining slugs
4. Close the lubricating drum and tumble for 15-20 min.

## Application recommendation

| Slug diameter | Amount of lubricant / 100 kg aluminum |
|---|---|
| 22 mm | 85 gram |
| 25 mm | 90 gram |
| 35 mm | 95 gram |
| 38 mm | 95 gram |
| 45 mm | 105 gram |
| 50 mm | 115 gram |
| 53 mm | 125 gram |
| 59 mm | 125 gram |
| 66 mm | 130 gram |

This recommendation is in general and could be adapted in the range of ± 15%

## Precautions

Observe the usual safety precautions when handling chemicals. The classification according to the legal regulations for transport, storage and handling of the product as well as further product-specific instructions are available in the EC-safety data sheet.

## Shelf Life

- In closed original packaging: 36 months
- Recommended storage temperature: 5°C - 30°C
- Avoid direct sunlight exposure

## Contact


Sapilub LTD
Bachrain 3
CH-8266 Steckborn
Switzerland

Phone +41 52 770 2120
Fax +41 52 761 2194
Mail sapilub@sapilub.ch
Net www.sapilub.ch

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**1.1 Product identifier:** LUBRIMET GR8

**Other means of identification:**

Not relevant

**1.2 Relevant identified uses of the substance or mixture and uses advised against:**

Relevant uses (Industrial user): Lubricant
For Industrial user only.

Uses advised against: All uses not specified in this section or in section 7.3

**1.3 Details of the supplier of the safety data sheet:**

Sapilub LTD
Bachrain 3
8266 Steckborn - Switzerland
Phone: +41 52 770 21 20
sapilub@sapilub.ch

**1.4 Emergency telephone number:** Tox Info Suisse, Zürich
+41 44 251 51 51
Non urgent inquiries: +41 44 251 66 66, info@toxinfo.ch

## SECTION 2: HAZARDS IDENTIFICATION

**2.1 Classification of the substance or mixture:**

**CLP Regulation (EC) No 1272/2008:**

Classification of this product has been carried out in accordance with CLP Regulation (EC) No 1272/2008.

Eye Irrit. 2: Eye irritation, Category 2, H319
STOT SE 3: Specific toxicity causing drowsiness and dizziness, single exposure, Category 3, H336

**2.2 Label elements:**

**CLP Regulation (EC) No 1272/2008:**

Warning

**Hazard statements:**

Eye Irrit. 2: H319 - Causes serious eye irritation.
STOT SE 3: H336 - May cause drowsiness or dizziness.

**Precautionary statements:**

P101: If medical advice is needed, have product container or label at hand.
P102: Keep out of reach of children.
P271: Use only outdoors or in a well-ventilated area.
P280: Wear protective gloves/protective clothing/respiratory protection/eye protection/protective footwear.
P304+P340: IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P305+P351+P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P403+P233: Store in a well-ventilated place. Keep container tightly closed.
P501: Dispose of contents/container according to the separated collection system used in your municipality.

**Substances that contribute to the classification**

propan-2-ol (CAS: 67-63-0)

**2.3 Other hazards:**

Product does not meet PBT/vPvB criteria
Endocrine-disrupting properties: The product does not meet the criteria.

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

**3.1 Substance:**

- CONTINUED ON NEXT PAGE -

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS (continued)

Not relevant

**3.2 Mixture:**

**Chemical description:** Mixture of substances

**Components:**

In accordance with Annex II of Regulation (EC) No 1907/2006 (point 3), the product contains:

| Identification | Chemical name/Classification | | | Concentration |
|---|---|---|---|---|
| CAS: 64-17-5<br>EC: 200-578-6<br>Index: 603-002-00-5<br>REACH: 01-2119457610-43-XXXX | **ethanol**[(1)] | | Self-classified | 20 - <40% |
| | Regulation 1272/2008 | Eye Irrit. 2: H319; Flam. Liq. 2: H225 - Danger | | |
| CAS: 67-63-0<br>EC: 200-661-7<br>Index: 603-117-00-0<br>REACH: 01-2119457558-25-XXXX | **propan-2-ol**[(1)] | | ATP CLP00 | 10 - <30% |
| | Regulation 1272/2008 | Eye Irrit. 2: H319; Flam. Liq. 2: H225; STOT SE 3: H336 - Danger | | |
| CAS: 9043-30-5<br>EC: 500-027-2<br>Index: Not relevant<br>REACH: Not relevant | **Isotridecanol, ethoxylated**[(1)] | | Self-classified | 1 - <3% |
| | Regulation 1272/2008 | Acute Tox. 4: H302; Eye Dam. 1: H318 - Danger | | |

[(1)] Substances presenting a health or environmental hazard which meet criteria laid down in Regulation (EU) No. 2020/878

To obtain more information on the hazards of the substances consult sections 11, 12 and 16.

**Other information:**

| Identification | Specific concentration limit |
|---|---|
| ethanol<br>CAS: 64-17-5<br>EC: 200-578-6 | % (w/w) >=50: Eye Irrit. 2 - H319 |

Acute toxicity estimate for the substance in Part 3 of Annex VI to Regulation (EC) No 1272/2008 or as determined in accordance with Annex I to that Regulation:

| Identification | Acute toxicity | | Genus |
|---|---|---|---|
| Isotridecanol, ethoxylated<br>CAS: 9043-30-5<br>EC: 500-027-2 | LD50 oral | 500 mg/kg | |
| | LD50 dermal | Not relevant | |
| | LC50 inhalation dust | Not relevant | |

## SECTION 4: FIRST AID MEASURES

**4.1 Description of first aid measures:**

The symptoms resulting from intoxication can appear after exposure, therefore, in case of doubt, seek medical attention for direct exposure to the chemical product or persistent discomfort, showing the SDS of this product.

**By inhalation:**

Remove the person affected from the area of exposure, provide with fresh air and keep at rest. In serious cases such as cardiorespiratory failure, artificial resuscitation techniques will be necessary (mouth to mouth resuscitation, cardiac massage, oxygen supply, etc.) requiring immediate medical assistance.

**By skin contact:**

In case of contact it is recommended to clean the affected area thoroughly with water and neutral soap. In case of changes to the skin (stinging, redness, rashes, blisters,…), seek medical advice with this Safety Data Sheet

**By eye contact:**

Rinse eyes thoroughly with lukewarm water for at least 15 minutes. Do not allow the person affected to rub or close their eyes. If the injured person uses contact lenses, these should be removed unless they are stuck to the eyes, in which case this could cause further damage. In all cases, after cleaning, a doctor should be consulted as quickly as possible with the SDS of the product.

**By ingestion/aspiration:**

Do not induce vomiting, but if it does happen keep the head down to avoid aspiration. Keep the person affected at rest. Rinse out the mouth and throat, as they may have been affected during ingestion.

- CONTINUED ON NEXT PAGE -

## SECTION 4: FIRST AID MEASURES (continued)

**4.2 Most important symptoms and effects, both acute and delayed:**

Acute and delayed effects are indicated in sections 2 and 11.

Drowsiness
Vertigo

**4.3 Indication of any immediate medical attention and special treatment needed:**

Treat symptomatically.
Forward this sheet to the doctor.

## SECTION 5: FIREFIGHTING MEASURES

**5.1 Extinguishing media:**

**Suitable extinguishing media:**

Foam extinguisher (AB), Dry Chemical Powder (ABC) Fire Extinguisher, Carbon dioxide extinguisher (BC)

**Unsuitable extinguishing media:**

Water jet

**5.2 Special hazards arising from the substance or mixture:**

As a result of combustion or thermal decomposition reactive sub-products are created that can become highly toxic and, consequently, can present a serious health risk.

**5.3 Advice for firefighters:**

Depending on the magnitude of the fire it may be necessary to use full protective clothing and Self Contained Breathing Apparatus. Minimum emergency facilities and equipment should be available (fire blankets, portable first aid kit,...)

**Additional provisions:**

Act in accordance with the Internal Emergency Plan and the Information Sheets on actions to take after an accident or other emergencies. Eliminate all sources of ignition. In case of fire, cool the storage containers and tanks for products susceptible to combustion, explosion or BLEVE as a result of high temperatures. Avoid spillage of the products used to extinguish the fire into an aqueous medium.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

**6.1 Personal precautions, protective equipment and emergency procedures:**

**For non-emergency personnel:**

Sweep up and shovel product or collect by other means and place in container for reuse (preferred) or disposal

**For emergency responders:**

Wear protective equipment. Keep unprotected persons away. See section 8.

**Other information:**

Ensure adequate ventilation.
Some risk of slipping due to spillage of product.

**6.2 Environmental precautions:**

This product is not classified as hazardous to the environment. Keep product away from drains, surface and ground water.

**6.3 Methods and material for containment and cleaning up:**

It is recommended:

Sweep up and shovel product or collect by other means and place in container for reuse (preferred) or disposal

**6.4 Reference to other sections:**

See sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

**7.1 Precautions for safe handling:**

A.- General precautions for safe use

- CONTINUED ON NEXT PAGE -

## SECTION 7: HANDLING AND STORAGE (continued)

Comply with the current legislation concerning the prevention of industrial risks. Keep containers hermetically sealed. Control spills and residues, destroying them with safe methods (section 6). Avoid leakages from the container. Maintain order and cleanliness where dangerous products are used.

B.- Technical recommendations for the prevention of fires and explosions

Avoid the evaporation of the product as it contains flammable substances, which could form flammable vapour/air mixtures in the presence of sources of ignition. Control sources of ignition (mobile phones, sparks,...) and transfer at slow speeds to avoid the creation of electrostatic charges. Consult section 10 for conditions and materials that should be avoided.

C.- Technical recommendations on general occupational hygiene

Do not eat or drink during the process, washing hands afterwards with suitable cleaning products.

D.- Technical recommendations to prevent environmental risks

It is recommended to have absorbent material available at close proximity to the product (See subsection 6.3)

**7.2 Conditions for safe storage, including any incompatibilities:**

A.- Specific storage requirements

Minimum Temp.: 5 ºC
Maximum Temp.: 30 ºC
Maximum time: 36 Months

B.- General conditions for storage

Avoid sources of heat, radiation, static electricity and contact with food. For additional information see subsection 10.5

**Other information:**

Keep only in original container.

Do not store with oxidizing or self-igniting materials.

Protect from heat/overheating.
Keep container in a well-ventilated place.
Keep container tighlty closed.

**7.3 Specific end use(s):**

Except for the instructions already specified it is not necessary to provide any special recommendation regarding the uses of this product.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

**8.1 Control parameters:**

Substances whose occupational exposure limits have to be monitored in the workplace (European OEL, not country-specific legislation):
Nuisance dust: Inhalable dust 10 mg/m3 // Respirable dust 4 mg/m3

**DNEL (Workers):**

| | | Short exposure | | Long exposure | |
|---|---|---|---|---|---|
| Identification | | Systemic | Local | Systemic | Local |
| propan-2-ol | Oral | Not relevant | Not relevant | Not relevant | Not relevant |
| CAS: 67-63-0 | Dermal | Not relevant | Not relevant | 888 mg/kg | Not relevant |
| EC: 200-661-7 | Inhalation | 1000 mg/m³ | Not relevant | 500 mg/m³ | Not relevant |
| ethanol | Oral | Not relevant | Not relevant | Not relevant | Not relevant |
| CAS: 64-17-5 | Dermal | Not relevant | Not relevant | 343 mg/kg | Not relevant |
| EC: 200-578-6 | Inhalation | Not relevant | Not relevant | 950 mg/m³ | Not relevant |

**DNEL (General population):**

- CONTINUED ON NEXT PAGE -

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION (continued)

| Identification | | Short exposure | | Long exposure | |
|---|---|---|---|---|---|
| | | Systemic | Local | Systemic | Local |
| propan-2-ol | Oral | 51 mg/kg | Not relevant | 26 mg/kg | Not relevant |
| CAS: 67-63-0 | Dermal | Not relevant | Not relevant | 319 mg/kg | Not relevant |
| EC: 200-661-7 | Inhalation | 178 mg/m³ | Not relevant | 114 mg/m³ | Not relevant |
| ethanol | Oral | Not relevant | Not relevant | 87 mg/kg | Not relevant |
| CAS: 64-17-5 | Dermal | Not relevant | Not relevant | 206 mg/kg | Not relevant |
| EC: 200-578-6 | Inhalation | Not relevant | Not relevant | 114 mg/m³ | Not relevant |

**PNEC:**

| Identification | | | | |
|---|---|---|---|---|
| propan-2-ol | STP | 2251 mg/L | Fresh water | 140,9 mg/L |
| CAS: 67-63-0 | Soil | 28 mg/kg | Marine water | 140,9 mg/L |
| EC: 200-661-7 | Intermittent | 140,9 mg/L | Sediment (Fresh water) | 552 mg/kg |
| | Oral | 0,16 g/kg | Sediment (Marine water) | 552 mg/kg |
| ethanol | STP | 580 mg/L | Fresh water | 0,96 mg/L |
| CAS: 64-17-5 | Soil | 0,63 mg/kg | Marine water | 0,79 mg/L |
| EC: 200-578-6 | Intermittent | 2,75 mg/L | Sediment (Fresh water) | 3,6 mg/kg |
| | Oral | 0,38 g/kg | Sediment (Marine water) | 2,9 mg/kg |

### 8.2 Exposure controls:

A.- Individual protection measures, such as personal protective equipment

As a preventative measure it is recommended to use basic Personal Protective Equipment, with the corresponding <<CE marking>> in accordance with Regulation (EU) 2016/425. For more information on Personal Protective Equipment (storage, use, cleaning, maintenance, class of protection,…) consult the information leaflet provided by the manufacturer. For more information see subsection 7.1. All information contained herein is a recommendation which needs some specification from the labour risk prevention services as it is not known whether the company has additional measures at its disposal.

B.- Respiratory protection

| Pictogram | PPE | Labelling | CEN Standard | Remarks |
|---|---|---|---|---|
| Mandatory respiratory tract protection | Filter mask for gases and vapours (Filter type: P2/FFP2) | CE CAT III | EN 405:2002+A1:2010 | Replace when there is a taste or smell of the contaminant inside the face mask. If the contaminant comes with warnings it is recommended to use isolation equipment. |

C.- Specific protection for the hands

| Pictogram | PPE | Labelling | CEN Standard | Remarks |
|---|---|---|---|---|
| Mandatory hand protection | Chemical protective gloves (Material: Linear low-density polyethylene (LLDPE), Breakthrough time: > 480 min, Thickness: 0.062 mm) | CE CAT III | EN ISO 21420:2020 | Replace the gloves at any sign of deterioration. |

As the product is a mixture of several substances, the resistance of the glove material can not be calculated in advance with total reliability and has therefore to be checked prior to the application.

D.- Eye and face protection

| Pictogram | PPE | Labelling | CEN Standard | Remarks |
|---|---|---|---|---|
| Mandatory face protection | Panoramic glasses against splash/projections. | CE CAT II | EN 166:2002<br>EN ISO 4007:2018 | Clean daily and disinfect periodically according to the manufacturer´s instructions. Use if there is a risk of splashing. |

E.- Body protection

- CONTINUED ON NEXT PAGE -

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION (continued)

| Pictogram | PPE | Labelling | CEN Standard | Remarks |
|---|---|---|---|---|
| | Work clothing | CE CAT I | | Replace before any evidence of deterioration. For periods of prolonged exposure to the product for professional/industrial users CE III is recommended, in accordance with the regulations in EN ISO 6529:2013, EN ISO 6530:2005, EN ISO 13688:2013, EN 464:1994. |
| | Anti-slip work shoes | CE CAT II | EN ISO 20347:2022 | Replace before any evidence of deterioration. For periods of prolonged exposure to the product for professional/industrial users CE III is recommended, in accordance with the regulations in EN ISO 20345:2022 y EN 13832-1:2019 |

F.- Additional emergency measures

It is advised to implement additional emergency equipments in workplaces that are particularly exposed to the product or in situations where risk assessments highlight the necessity of such equipments.

| Emergency measure | Standards | Emergency measure | Standards |
|---|---|---|---|
| Emergency shower | ANSI Z358-1<br>ISO 3864-1:2011, ISO 3864-4:2011 | Eyewash stations | DIN 12 899<br>ISO 3864-1:2011, ISO 3864-4:2011 |

**Environmental exposure controls:**

To comply with environmental protection regulations, it is recommended to prevent any spillage of the product and its container. For more detailed information, please refer to subsection 7.1.D.

**Volatile organic compounds:**

With regard to Directive 2010/75/EU, this product has the following characteristics:

| | |
|---|---|
| V.O.C. (Supply): | 45,61 % weight |
| V.O.C. density at 20 ºC: | 387,72 kg/m³ (387,72 g/L) |
| Average carbon number: | 2,55 |
| Average molecular weight: | 53,85 g/mol |

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

**9.1 Information on basic physical and chemical properties:**

For complete information see the product datasheet.

**Appearance:**

| | |
|---|---|
| Physical state at 20 ºC: | Solid |
| Appearance: | Paste |
| Colour: | White |
| Odour: | Alcohol |
| Odour threshold: | Not relevant * |

**Volatility:**

| | |
|---|---|
| Boiling point at atmospheric pressure: | Not relevant * |
| Vapour pressure at 20 ºC: | Not relevant * |
| Vapour pressure at 50 ºC: | Not relevant * |
| Evaporation rate at 20 ºC: | Not relevant * |

**Product description:**

| | |
|---|---|
| Density at 20 ºC: | 830 - 870 kg/m³ |
| Relative density at 20 ºC: | Not relevant * |
| Dynamic viscosity at 20 ºC: | Not relevant * |
| Kinematic viscosity at 20 ºC: | Not relevant * |

*Not relevant due to the nature of the product, not providing information property of its hazards.

- CONTINUED ON NEXT PAGE -

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES (continued)

| | |
|---|---|
| Kinematic viscosity at 40 ºC: | Not relevant * |
| Concentration: | Not relevant * |
| pH: | Not relevant * |
| Vapour density at 20 ºC: | Not relevant * |
| Partition coefficient n-octanol/water 20 ºC: | Not relevant * |
| Solubility in water at 20 ºC: | Not relevant * |
| Solubility properties: | Immiscible |
| Decomposition temperature: | Not relevant * |
| Melting point/freezing point: | Not relevant * |
| **Flammability:** | |
| Flash Point: | 10 - 13 ºC |
| Flammability (solid, gas): | Not relevant * |
| Autoignition temperature: | Not relevant * |
| Lower flammability limit: | Not relevant * |
| Upper flammability limit: | Not relevant * |
| **Explosive (Solid):** | |
| Lower explosive limit: | Not relevant * |
| Upper explosive limit: | Not relevant * |
| **Particle characteristics:** | |
| Median equivalent diameter: | Not relevant * |

**9.2 Other information:**

| | |
|---|---|
| **Information with regard to physical hazard classes:** | |
| Explosive properties: | Not relevant * |
| Oxidising properties: | Not relevant * |
| Corrosive to metals: | Not relevant * |
| Heat of combustion: | Not relevant * |
| Aerosols-total percentage (by mass) of flammable components: | Not relevant * |
| **Other safety characteristics:** | |
| Surface tension at 20 ºC: | Not relevant * |
| Refraction index: | Not relevant * |

Burning time (100 mm): 20 s

*Not relevant due to the nature of the product, not providing information property of its hazards.

## SECTION 10: STABILITY AND REACTIVITY

**10.1 Reactivity:**

No hazardous reactions are expected because the product is stable under recommended storage conditions. See section 7 from Safety Data Sheet.

**10.2 Chemical stability:**

Chemically stable under the indicated conditions of storage, handling and use.

**10.3 Possibility of hazardous reactions:**

Under the specified conditions, hazardous reactions that lead to excessive temperatures or pressure are not expected.

**10.4 Conditions to avoid:**

Applicable for handling and storage at room temperature:

- CONTINUED ON NEXT PAGE -

## SECTION 10: STABILITY AND REACTIVITY (continued)

| Shock and friction | Contact with air | Increase in temperature | Sunlight | Humidity |
|---|---|---|---|---|
| Not applicable | Not applicable | Precaution | Precaution | Not applicable |

**10.5 Incompatible materials:**

| Acids | Water | Oxidising materials | Combustible materials | Others |
|---|---|---|---|---|
| Avoid strong acids | Not applicable | Avoid direct impact | Not applicable | Avoid alkalis or strong bases |

**10.6 Hazardous decomposition products:**

See subsection 10.3, 10.4 and 10.5 to find out the specific decomposition products. Depending on the decomposition conditions, complex mixtures of chemical substances can be released: carbon dioxide ($CO_2$), carbon monoxide and other organic compounds.

## SECTION 11: TOXICOLOGICAL INFORMATION

**11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008:**

The experimental information related to the toxicological properties of the product itself is not available

**Dangerous health implications:**

In case of exposure that is repetitive, prolonged or at concentrations higher than the recommended occupational exposure limits, adverse effects on health may result, depending on the means of exposure:

A- Ingestion (acute effect):

- Acute toxicity: Based on available data, the classification criteria are not met, however, it contains substances classified as dangerous for consumption. For more information see section 3.
- Corrosivity/Irritability: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.

B- Inhalation (acute effect):

- Acute toxicity : Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for inhalation. For more information see section 3.
- Corrosivity/Irritability: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.

C- Contact with the skin and the eyes (acute effect):

- Contact with the skin: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for skin contact. For more information see section 3.
- Contact with the eyes: Produces eye damage after contact.

D- CMR effects (carcinogenicity, mutagenicity and toxicity to reproduction):

- Carcinogenicity: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for the effects mentioned. For more information see section 3.
  IARC: ethanol (1); propan-2-ol (3); White mineral oil, 7 < v <=20.5mm2/s (40ºC) (3)
- Mutagenicity: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.
- Reproductive toxicity: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.

E- Sensitizing effects:

- Respiratory: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous with sensitising effects. For more information see section 3.
- Skin: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.

F- Specific target organ toxicity (STOT) - single exposure:

Exposure in high concentration can interfere with the central nervous system causing headache, dizziness, vertigo, nausea, vomiting, confusion, and in serious cases, loss of consciousness.

G- Specific target organ toxicity (STOT)-repeated exposure:

- CONTINUED ON NEXT PAGE -

## SECTION 11: TOXICOLOGICAL INFORMATION (continued)

- Specific target organ toxicity (STOT)-repeated exposure: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.
- Skin: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.

H- Aspiration hazard:

Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.

**Other information:**

Not relevant

**Specific toxicology information on the substances:**

| Identification | Acute toxicity | | Genus |
|---|---|---|---|
| Isotridecanol, ethoxylated<br>CAS: 9043-30-5<br>EC: 500-027-2 | LD50 oral | 500 mg/kg | |
| | LD50 dermal | >2000 mg/kg | |
| | LC50 inhalation dust | >5 mg/L | |
| ethanol<br>CAS: 64-17-5<br>EC: 200-578-6 | LD50 oral | 6200 mg/kg | Rat |
| | LD50 dermal | 20000 mg/kg | Rabbit |
| | LC50 inhalation vapour | 124,7 mg/L (4 h) | Rat |
| propan-2-ol<br>CAS: 67-63-0<br>EC: 200-661-7 | LD50 oral | >5840 mg/kg | Rat |
| | LD50 dermal | >13900 mg/kg | Rabbit |
| | LC50 inhalation vapour | >25 mg/L (6 h) | Rat |

**Acute Toxicity Estimate (ATE mix):**

| ATE mix | | Ingredient(s) of unknown toxicity |
|---|---|---|
| Oral | 47619,05 mg/kg (Calculation method) | 0 % |
| Dermal | >2000 mg/kg (Calculation method) | 0 % |
| LC50 inhalation dust | >5 mg/L (4 h) (Calculation method) | 0 % |

**11.2 Information on other hazards:**

**Endocrine disrupting properties**

Endocrine-disrupting properties: The product does not meet the criteria.

**Other information**

Not relevant

## SECTION 12: ECOLOGICAL INFORMATION

The experimental information related to the eco-toxicological properties of the product itself is not available

Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.

**12.1 Toxicity:**

**Acute toxicity:**

| Identification | Concentration | | Species | Genus |
|---|---|---|---|---|
| propan-2-ol<br>CAS: 67-63-0<br>EC: 200-661-7 | LC50 | 9640 mg/L (96 h) | Pimephales promelas | Fish |
| | EC50 | 10000 mg/L (24 h) | Daphnia magna | Crustacean |
| | EC50 | Not relevant | | |
| ethanol<br>CAS: 64-17-5<br>EC: 200-578-6 | LC50 | 11000 mg/L (96 h) | Alburnus alburnus | Fish |
| | EC50 | 9268 mg/L (48 h) | Daphnia magna | Crustacean |
| | EC50 | 1450 mg/L (192 h) | Microcystis aeruginosa | Algae |

**Chronic toxicity:**

| Identification | Concentration | | Species | Genus |
|---|---|---|---|---|
| ethanol<br>CAS: 64-17-5 EC: 200-578-6 | NOEC | 250 mg/L | Danio rerio | Fish |
| | NOEC | 2 mg/L | Ceriodaphnia dubia | Crustacean |

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## SECTION 12: ECOLOGICAL INFORMATION (continued)

**12.2 Persistence and degradability:**

**Substance-specific information:**

| Identification | Degradability | | Biodegradability | |
|---|---|---|---|---|
| propan-2-ol<br>CAS: 67-63-0<br>EC: 200-661-7 | BOD5 | 1,19 g O2/g | Concentration | 100 mg/L |
| | COD | 2,23 g O2/g | Period | 14 days |
| | BOD5/COD | 0,53 | % Biodegradable | 86 % |
| ethanol<br>CAS: 64-17-5<br>EC: 200-578-6 | BOD5 | Not relevant | Concentration | 100 mg/L |
| | COD | Not relevant | Period | 14 days |
| | BOD5/COD | Not relevant | % Biodegradable | 89 % |

**12.3 Bioaccumulative potential:**

**Substance-specific information:**

| Identification | Bioaccumulation potential | |
|---|---|---|
| propan-2-ol<br>CAS: 67-63-0<br>EC: 200-661-7 | BCF | 3 |
| | Pow Log | 0.05 |
| | Potential | Low |
| ethanol<br>CAS: 64-17-5<br>EC: 200-578-6 | BCF | 3 |
| | Pow Log | -0.31 |
| | Potential | Low |

**12.4 Mobility in soil:**

| Identification | Absorption/desorption | | Volatility | |
|---|---|---|---|---|
| propan-2-ol<br>CAS: 67-63-0<br>EC: 200-661-7 | Koc | 1.5 | Henry | 8,207E-1 Pa·m³/mol |
| | Conclusion | Very High | Dry soil | Yes |
| | Surface tension | 2,24E-2 N/m (25 ºC) | Moist soil | Yes |
| ethanol<br>CAS: 64-17-5<br>EC: 200-578-6 | Koc | 1 | Henry | 4,61E-1 Pa·m³/mol |
| | Conclusion | Very High | Dry soil | Yes |
| | Surface tension | 2,339E-2 N/m (25 ºC) | Moist soil | Yes |

**12.5 Results of PBT and vPvB assessment:**

Product does not meet PBT/vPvB criteria

**12.6 Endocrine disrupting properties:**

Endocrine-disrupting properties: The product does not meet the criteria.

**12.7 Other adverse effects:**

Not described

## SECTION 13: DISPOSAL CONSIDERATIONS

**13.1 Waste treatment methods:**

| Code | Description | Waste class (Regulation (EU) No 1357/2014) |
|---|---|---|
| 12 01 12* | spent waxes and fats | Hazardous |

**Type of waste (Regulation (EU) No 1357/2014):**

HP5 Specific Target Organ Toxicity (STOT)/Aspiration Toxicity, HP4 Irritant — skin irritation and eye damage

**Waste management (disposal and evaluation):**

Consult the authorized waste service manager on the assessment and disposal operations in accordance with Annex 1 and Annex 2 (Directive 2008/98/EC). As under 15 01 (2014/955/EC) of the code and in case the container has been in direct contact with the product, it will be processed the same way as the actual product. Otherwise, it will be processed as non-hazardous residue. Waste should not be disposed of to drains. See paragraph 6.2.

**Regulations related to waste management:**

In accordance with Annex II of Regulation (EC) No 1907/2006 (REACH) the community or state provisions related to waste management are stated

Community legislation: Directive 2008/98/EC, 2014/955/EU, Regulation (EU) No 1357/2014

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## SECTION 14: TRANSPORT INFORMATION

**Transport of dangerous goods by land:**

With regard to ADR 2023 and RID 2023:


| | | |
|---|---|---|
| **14.1** | **UN number or ID number:** | UN3175 |
| **14.2** | **UN proper shipping name:** | SOLIDS CONTAINING FLAMMABLE LIQUID, N.O.S. (ethanol; propan-2-ol) |
| **14.3** | **Transport hazard class(es):** | 4.1 |
| | Labels: | 4.1 |
| **14.4** | **Packing group:** | II |
| **14.5** | **Environmental hazards:** | No |
| **14.6** | **Special precautions for user** | |
| | Special regulations: | 216, 274, 601 |
| | Tunnel restriction code: | E |
| | Physico-Chemical properties: | see section 9 |
| | Limited quantities: | 1 kg |
| **14.7** | **Maritime transport in bulk according to IMO instruments:** | Not relevant |

**Transport of dangerous goods by sea:**

With regard to IMDG 41-22:


| | | |
|---|---|---|
| **14.1** | **UN number or ID number:** | UN3175 |
| **14.2** | **UN proper shipping name:** | SOLIDS CONTAINING FLAMMABLE LIQUID, N.O.S. (ethanol; propan-2-ol) |
| **14.3** | **Transport hazard class(es):** | 4.1 |
| | Labels: | 4.1 |
| **14.4** | **Packing group:** | II |
| **14.5** | **Marine pollutant:** | No |
| **14.6** | **Special precautions for user** | |
| | Special regulations: | 274, 216 |
| | EmS Codes: | F-A, S-I |
| | Physico-Chemical properties: | see section 9 |
| | Limited quantities: | 1 kg |
| | Segregation group: | Not relevant |
| **14.7** | **Maritime transport in bulk according to IMO instruments:** | Not relevant |

**Transport of dangerous goods by air:**

With regard to IATA/ICAO 2025:


| | | |
|---|---|---|
| **14.1** | **UN number or ID number:** | UN3175 |
| **14.2** | **UN proper shipping name:** | SOLIDS CONTAINING FLAMMABLE LIQUID, N.O.S. (ethanol; propan-2-ol) |
| **14.3** | **Transport hazard class(es):** | 4.1 |
| | Labels: | 4.1 |
| **14.4** | **Packing group:** | II |
| **14.5** | **Environmental hazards:** | No |
| **14.6** | **Special precautions for user** | |
| | Physico-Chemical properties: | see section 9 |
| **14.7** | **Maritime transport in bulk according to IMO instruments:** | Not relevant |

## SECTION 15: REGULATORY INFORMATION

**15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture:**

- CONTINUED ON NEXT PAGE -

## SECTION 15: REGULATORY INFORMATION (continued)

- Article 95, REGULATION (EU) No 528/2012: *propan-2-ol (67-63-0) - PT: (1,2,4)* ; *ethanol (64-17-5) - PT: (1,2,4,6)*
- Candidate substances for authorisation under the Regulation (EC) No 1907/2006 (REACH): Not relevant
- Regulation (EU) 2019/1021 on persistent organic pollutants: Not relevant
- Regulation (EU) No 2024/590, about substances that deplete the ozone layer: Not relevant
- REGULATION (EU) No 649/2012, in relation to the import and export of hazardous chemical products: Not relevant
- Substances included in Annex XIV of REACH ("Authorisation List") and sunset date: Not relevant

**Seveso III:**

Not relevant

**Limitations to commercialisation and the use of certain dangerous substances and mixtures (Annex XVII REACH, etc ….):**

Not relevant

**Specific provisions in terms of protecting people or the environment:**

It is recommended to use the information included in this safety data sheet as a basis for conducting workplace-specific risk assessments in order to establish the necessary risk prevention measures for the handling, use, storage and disposal of this product.

**Other legislation:**

The product could be affected by sectorial legislation

**15.2 Chemical safety assessment:**

The supplier has not carried out evaluation of chemical safety.

## SECTION 16: OTHER INFORMATION

**Legislation related to safety data sheets:**

The SDS shall be supplied in an official language of the country where the product is placed on the market. This safety data sheet has been designed in accordance with ANNEX II-Guide to the compilation of safety data sheets of Regulation (EC) No 1907/2006 (COMMISSION REGULATION (EU) 2020/878).

**Modifications related to the previous Safety Data Sheet which concerns the ways of managing risks.:**

Not relevant

**Texts of the legislative phrases mentioned in section 2:**

H319: Causes serious eye irritation.
H336: May cause drowsiness or dizziness.

**Texts of the legislative phrases mentioned in section 3:**

The phrases indicated do not refer to the product itself; they are present merely for informative purposes and refer to the individual components which appear in section 3

**CLP Regulation (EC) No 1272/2008:**

Acute Tox. 4: H302 - Harmful if swallowed.
Eye Dam. 1: H318 - Causes serious eye damage.
Eye Irrit. 2: H319 - Causes serious eye irritation.
Flam. Liq. 2: H225 - Highly flammable liquid and vapour.
STOT SE 3: H336 - May cause drowsiness or dizziness.

**Classification procedure:**

Eye Irrit. 2: Calculation method
STOT SE 3: Calculation method

**Advice related to training:**

Training is recommended in order to prevent industrial risks for staff using this product and to facilitate their comprehension and interpretation of this safety data sheet, as well as the label on the product.

**Principal bibliographical sources:**

http://echa.europa.eu
http://eur-lex.europa.eu

**Abbreviations and acronyms:**

- CONTINUED ON NEXT PAGE -

## SECTION 16: OTHER INFORMATION (continued)

ADR: European agreement concerning the international carriage of dangerous goods by road
IMDG: International maritime dangerous goods code
IATA: International Air Transport Association
ICAO: International Civil Aviation Organisation
COD: Chemical Oxygen Demand
BOD5: 5day biochemical oxygen demand
BCF: Bioconcentration factor
LD50: Lethal Dose 50
LC50: Lethal Concentration 50
EC50: Effective concentration 50
LogPOW: Octanolwater partition coefficient
Koc: Partition coefficient of organic carbon
UFI: unique formula identifier
IARC: International Agency for Research on Cancer



- END OF SAFETY DATA SHEET -

SAPILUB

# ALULIQUID 13

## Metal cleaner

# TECHNICAL DATA SHEET

## Product Information

**ALULIQUID 13** is a liquid phosphate and surfactant- free alkaline metal cleaner, which achieves an excellent result even at a low bath concentration.

**ALULIQUID 13** metal cleaner results in bright and shiny aluminum appearance by not etching the aluminum surface.

## Field of Application

**ALULIQUID 13** is a non-foaming cleaner for aluminum aerosol cans, developed for spray washing machines.

## Typical Properties

| | |
|---|---|
| **Appearance:** | blue |
| **Physical aspect:** | liquid |
| **pH-Value 1% in $H_2O$**: | 12.2 @ 20°C |
| **pH-Value 1% in $H_2O$**: | 10.4 @ 70°C (bath) |
| **Density at 20°C:** | 1.30 g/cm³ |
| **Composition:** | mix of alkaline and inhibitors |

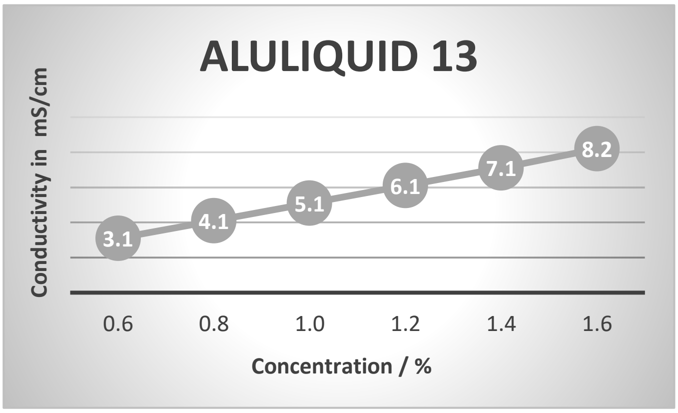
## Performance Properties

| | |
|---|---|
| **Temperature / bath:** | 70°C ± 5°C |
| **Conductivity / bath:** | 3.5 - 7.0 mS/cm |
| **Concentration:** | 0.7 - 1.4 % |

## Sewage / Environment

Bath solutions, rinsing water and concentrates must be treated before being discharged into the sewage system in accordance with local discharge regulations.

## Diagram Concentration / Conductivity




## Precautions

Observe the usual safety precautions when handling chemicals. The classification according to the legal regulations for transport, storage and handling of the product as well as further product-specific instructions are available in the EC-safety data sheet.

## Shelf Life

In closed original packaging: 36 months

Recommended storage temperature: 5°C - 35°C

Avoid direct sunlight exposure

## Contact

SAPILUB LTD SWITZERLAND

Sapilub LTD
Bachrain 3
CH-8266 Steckborn
Switzerland

Phone +41 52 770 2120
Fax +41 52 761 2194
Mail sapilub@sapilub.ch
Net www.sapilub.ch

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**1.1 Product identifier:** ALULIQUID 13

**Other means of identification:**

Non-applicable

**1.2 Relevant identified uses of the substance or mixture and uses advised against:**

Relevant uses: Detergent. For industrial user only.

Cleaning of aluminium surfaces.

Uses advised against: All uses not specified in this section or in section 7.3

**1.3 Details of the supplier of the safety data sheet:**

Sapilub LTD
Bachrain 3
8266 Steckborn - Switzerland
Phone: +41 52 770 21 20
sapilub@sapilub.ch

**1.4 Emergency telephone number:** Tox Info Suisse, Zürich
+41 44 251 51 51
Non urgent inquiries: +41 44 251 66 66, info@toxinfo.ch

## SECTION 2: HAZARDS IDENTIFICATION

**2.1 Classification of the substance or mixture:**

**CLP Regulation (EC) No 1272/2008:**

Classification of this product has been carried out in accordance with CLP Regulation (EC) No 1272/2008.

Eye Dam. 1: Serious eye damage, Category 1, H318
Met. Corr. 1: Corrosive to metals, Category 1, H290
Skin Corr. 1: Skin corrosion, Category 1, H314

**2.2 Label elements:**

**CLP Regulation (EC) No 1272/2008:**

**Danger**

**Hazard statements:**

Met. Corr. 1: H290 - May be corrosive to metals.
Skin Corr. 1: H314 - Causes severe skin burns and eye damage.

**Precautionary statements:**

P101: If medical advice is needed, have product container or label at hand.
P102: Keep out of reach of children.
P280: Wear protective gloves/face protection/protective clothing/protective footwear.
P301+P330+P331: IF SWALLOWED: Rinse mouth. Do NOT induce vomiting.
P303+P361+P353: IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower.
P304+P340: IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P305+P351+P338: IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P501: Dispose of contents/container according to the separated collection system used in your municipality.

**Substances that contribute to the classification**

potassium hydroxide (CAS: 1310-58-3)

**2.3 Other hazards:**

Product fails to meet PBT/vPvB criteria
Endocrine-disrupting properties: The product fails to meet the criteria.

- CONTINUED ON NEXT PAGE -

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

**3.1 Substance:**

Non-applicable

**3.2 Mixture:**

**Chemical description:** Mixture of substances

**Components:**

In accordance with Annex II of Regulation (EC) No 1907/2006 (point 3), the product contains:

| Identification | Chemical name/Classification | | Concentration |
|---|---|---|---|
| CAS: 1310-58-3<br>EC: 215-181-3<br>Index: 019-002-00-8<br>REACH: 01-2119487136-33-XXXX | **potassium hydroxide**[1] | Self-classified | **5 - <15 %** |
| | Regulation 1272/2008 | Acute Tox. 4: H302; Met. Corr. 1: H290; Skin Corr. 1A: H314 - Danger | |
| CAS: 5064-31-3<br>EC: 225-768-6<br>Index: 607-620-00-6<br>REACH: 01-2119519239-36-XXXX | **trisodium nitrilotriacetate**[1] | ATP ATP01 | **1 - <5 %** |
| | Regulation 1272/2008 | Acute Tox. 4: H302; Carc. 2: H351; Eye Irrit. 2: H319 - Warning | |

[1] Substances presenting a health or environmental hazard which meet criteria laid down in Regulation (EU) No. 2020/878

To obtain more information on the hazards of the substances consult sections 11, 12 and 16.

**Other information:**

| Identification | Specific concentration limit |
|---|---|
| potassium hydroxide<br>CAS: 1310-58-3<br>EC: 215-181-3 | % (w/w) >=5: Skin Corr. 1A - H314<br>2<= % (w/w) <5: Skin Corr. 1B - H314<br>0.5<= % (w/w) <2: Skin Irrit. 2 - H315<br>% (w/w) >=0.5: Eye Irrit. 2 - H319 |
| trisodium nitrilotriacetate<br>CAS: 5064-31-3<br>EC: 225-768-6 | % (w/w) >=5: Carc. 2 - H351 |

## SECTION 4: FIRST AID MEASURES

**4.1 Description of first aid measures:**

Request medical assistance immediately, showing the SDS of this product.

**By inhalation:**

This product does not contain substances classified as hazardous for inhalation, however, in case of symptoms of intoxication remove the person affected from the exposure area and provide with fresh air. Seek medical attention if the symptoms get worse or persist.

**By skin contact:**

Remove contaminated clothing and footwear, rinse skin or shower the person affected if appropriate with plenty of cold water and neutral soap. In serious cases see a doctor. If the product causes burns or freezing, clothing should not be removed as this could worsen the injury caused if it is stuck to the skin. If blisters form on the skin, these should never be burst as this will increase the risk of infection.

**By eye contact:**

Rinse eyes thoroughly with lukewarm water for at least 15 minutes. Do not allow the person affected to rub or close their eyes. If the injured person uses contact lenses, these should be removed unless they are stuck to the eyes, in which case this could cause further damage. In all cases, after cleaning, a doctor should be consulted as quickly as possible with the SDS of the product.

**By ingestion/aspiration:**

Request immediate medical assistance, showing the SDS of this product. Do not induce vomiting, because its expulsion from the stomach can be hazardous to the mucus of the main digestive tract, and also risk damage to the respiratory system through inhalation. Rinse out the mouth and throat, as they may have been affected during ingestion. In the case of loss of consciousness do not administer anything orally unless supervised by a doctor. Keep the person affected at rest.

Change soaked clothing immediatly

- CONTINUED ON NEXT PAGE -

## SECTION 4: FIRST AID MEASURES (continued)

**4.2 Most important symptoms and effects, both acute and delayed:**

Acute and delayed effects are indicated in sections 2 and 11.

Causes chemical burns.

**4.3 Indication of any immediate medical attention and special treatment needed:**

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

**5.1 Extinguishing media:**

**Suitable extinguishing media:**

Product is non-flammable under normal conditions of storage, handling and use. In the case of combustion as a result of improper handling, storage or use preferably use polyvalent powder extinguishers (ABC powder), in accordance with the Regulation on fire protection systems.

**Unsuitable extinguishing media:**

Non-applicable

**5.2 Special hazards arising from the substance or mixture:**

As a result of combustion or thermal decomposition reactive sub-products are created that can become highly toxic and, consequently, can present a serious health risk.

**5.3 Advice for firefighters:**

Depending on the magnitude of the fire it may be necessary to use full protective clothing and self-contained breathing apparatus (SCBA). Minimum emergency facilities and equipment should be available (fire blankets, portable first aid kit,...) in accordance with Directive 89/654/EC.

**Additional provisions:**

Act in accordance with the Internal Emergency Plan and the Information Sheets on actions to take after an accident or other emergencies. Eliminate all sources of ignition. In case of fire, cool the storage containers and tanks for products susceptible to combustion, explosion or BLEVE as a result of high temperatures. Avoid spillage of the products used to extinguish the fire into an aqueous medium.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

**6.1 Personal precautions, protective equipment and emergency procedures:**

**For non-emergency personnel:**

Isolate leaks provided that there is no additional risk for the people performing this task. Personal protection equipment must be used against potential contact with the spilt product (See section 8). Evacuate the area and keep out those who do not have protection.

**For emergency responders:**

Wear protective equipment. Keep unprotected persons away. See section 8.

**Other information:**

High risk of slipping due to leakage/spillage of product

**6.2 Environmental precautions:**

This product is not classified as hazardous to the environment. Keep product away from drains, surface and ground water.

**6.3 Methods and material for containment and cleaning up:**

It is recommended:

Absorb the spillage using sand or inert absorbent and move it to a safe place. Do not absorb in sawdust or other combustible absorbents. For any concern related to disposal consult section 13.

**6.4 Reference to other sections:**

See sections 8 and 13.

- CONTINUED ON NEXT PAGE -

## SECTION 7: HANDLING AND STORAGE

**7.1 Precautions for safe handling:**

A.- General precautions for safe use

Comply with the current legislation concerning the prevention of industrial risks. Control spills and residues, destroying them with safe methods (section 6). Avoid leakages from the container. Maintain order and cleanliness where dangerous products are used. KEEP ONLY IN ORIGINAL PACKAGING.

B.- Technical recommendations for the prevention of fires and explosions

Product is non-flammable under normal conditions of storage, handling and use. It is recommended to transfer at slow speeds to avoid the generation of electrostatic charges that can affect flammable products. Consult section 10 for information on conditions and materials that should be avoided.

C.- Technical recommendations on general occupational hygiene

Do not eat or drink during the process, washing hands afterwards with suitable cleaning products.

D.- Technical recommendations to prevent environmental risks

It is recommended to have absorbent material available at close proximity to the product (See subsection 6.3)

**7.2 Conditions for safe storage, including any incompatibilities:**

A.- Technical measures for storage

Minimum Temp.: 5 ºC

Maximum Temp.: 30 ºC

Maximum time: 36 Months

B.- General conditions for storage

Avoid sources of heat, radiation, static electricity and contact with food. For additional information see subsection 10.5

**Other information:**

Keep only in tightly closed original container.
Provide alkali-resistant floor.
Do not store together with acids.
Do not keep at temperatures below -3°C.

**7.3 Specific end use(s):**

Except for the instructions already specified it is not necessary to provide any special recommendation regarding the uses of this product.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

**8.1 Control parameters:**

Substances whose occupational exposure limits have to be monitored in the workplace (European OEL, not country-specific legislation):
There are no applicable occupational exposure limits for the substances contained in the product

**DNEL (Workers):**

| | | Short exposure | | Long exposure | |
|---|---|---|---|---|---|
| Identification | | Systemic | Local | Systemic | Local |
| potassium hydroxide | Oral | Non-applicable | Non-applicable | Non-applicable | Non-applicable |
| CAS: 1310-58-3 | Dermal | Non-applicable | Non-applicable | Non-applicable | Non-applicable |
| EC: 215-181-3 | Inhalation | Non-applicable | Non-applicable | Non-applicable | 1 mg/m³ |
| trisodium nitrilotriacetate | Oral | Non-applicable | Non-applicable | Non-applicable | Non-applicable |
| CAS: 5064-31-3 | Dermal | Non-applicable | Non-applicable | Non-applicable | Non-applicable |
| EC: 225-768-6 | Inhalation | 9,6 mg/m³ | Non-applicable | 3,2 mg/m³ | Non-applicable |

**DNEL (General population):**

- CONTINUED ON NEXT PAGE -

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION (continued)

| Identification | | Short exposure | | Long exposure | |
|---|---|---|---|---|---|
| | | Systemic | Local | Systemic | Local |
| potassium hydroxide | Oral | Non-applicable | Non-applicable | Non-applicable | Non-applicable |
| CAS: 1310-58-3 | Dermal | Non-applicable | Non-applicable | Non-applicable | Non-applicable |
| EC: 215-181-3 | Inhalation | Non-applicable | Non-applicable | Non-applicable | 1 mg/m³ |
| trisodium nitrilotriacetate | Oral | 0,9 mg/kg | Non-applicable | 0,3 mg/kg | Non-applicable |
| CAS: 5064-31-3 | Dermal | Non-applicable | Non-applicable | Non-applicable | Non-applicable |
| EC: 225-768-6 | Inhalation | 2,4 mg/m³ | Non-applicable | 0,8 mg/m³ | Non-applicable |

**PNEC:**

| Identification | | | | |
|---|---|---|---|---|
| trisodium nitrilotriacetate | STP | 270 mg/L | Fresh water | 0,93 mg/L |
| CAS: 5064-31-3 | Soil | Non-applicable | Marine water | 0,093 mg/L |
| EC: 225-768-6 | Intermittent | 0,8 mg/L | Sediment (Fresh water) | Non-applicable |
| | Oral | Non-applicable | Sediment (Marine water) | Non-applicable |

**8.2 Exposure controls:**

A.- Individual protection measures, such as personal protective equipment

As a preventative measure it is recommended to use basic Personal Protective Equipment, with the corresponding <<CE marking>> in accordance with Regulation (EU) 2016/425. For more information on Personal Protective Equipment (storage, use, cleaning, maintenance, class of protection,…) consult the information leaflet provided by the manufacturer. For more information see subsection 7.1. All information contained herein is a recommendation which needs some specification from the labour risk prevention services as it is not known whether the company has additional measures at its disposal.

B.- Respiratory protection

The use of protection equipment will be necessary if a mist forms or if the occupational exposure limits are exceeded.

C.- Specific protection for the hands

| Pictogram | PPE | Labelling | CEN Standard | Remarks |
|---|---|---|---|---|
| Mandatory hand protection | Chemical protective gloves (Material: Linear low-density polyethylene (LLDPE), Breakthrough time: > 480 min, Thickness: 0.062 mm) | CE CAT III | EN ISO 21420:2020 | Replace the gloves at any sign of deterioration. |

As the product is a mixture of several substances, the resistance of the glove material can not be calculated in advance with total reliability and has therefore to be checked prior to the application.

D.- Eye and face protection

| Pictogram | PPE | Labelling | CEN Standard | Remarks |
|---|---|---|---|---|
| Mandatory face protection | Face shield | CE CAT II | EN 166:2002<br>EN 167:2002<br>EN 168:2002<br>EN ISO 4007:2018 | Clean daily and disinfect periodically according to the manufacturer´s instructions. Use if there is a risk of splashing. |

E.- Body protection

| Pictogram | PPE | Labelling | CEN Standard | Remarks |
|---|---|---|---|---|
| Mandatory complete body protection | Disposable clothing for protection against chemical risks | CE CAT III | EN 13034:2005+A1:2009<br>EN 168:2002<br>EN ISO 13982-1:2004/A1:2010<br>EN ISO 6529:2013<br>EN ISO 6530:2005<br>EN 464:1994 | For professional use only. Clean periodically according to the manufacturer´s instructions. |
| Mandatory foot protection | Safety footwear for protection against chemical risk | CE CAT III | EN ISO 20345:2011<br>EN 13832-1:2019 | Replace boots at any sign of deterioration. |

F.- Additional emergency measures

- CONTINUED ON NEXT PAGE -

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION (continued)

| Emergency measure | Standards | Emergency measure | Standards |
|---|---|---|---|
| Emergency shower | ANSI Z358-1<br>ISO 3864-1:2011, ISO 3864-4:2011 | Eyewash stations | DIN 12 899<br>ISO 3864-1:2011, ISO 3864-4:2011 |

**Environmental exposure controls:**

In accordance with the community legislation for the protection of the environment it is recommended to avoid environmental spillage of both the product and its container. For additional information see subsection 7.1.D

**Volatile organic compounds:**

With regard to Directive 2010/75/EU, this product has the following characteristics:

| | |
|---|---|
| V.O.C. (Supply): | 0 % weight |
| V.O.C. density at 20 ºC: | 0 kg/m³ (0 g/L) |
| Average carbon number: | Non-applicable |
| Average molecular weight: | Non-applicable |

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1 **Information on basic physical and chemical properties:**

For complete information see the product datasheet.

**Appearance:**

| | |
|---|---|
| Physical state at 20 ºC: | Liquid |
| Appearance: | Transparent |
| Colour: | Blue |
| Odour: | Characteristic |
| Odour threshold: | Non-applicable * |

**Volatility:**

| | |
|---|---|
| Boiling point at atmospheric pressure: | Non-applicable * |
| Vapour pressure at 20 ºC: | Non-applicable * |
| Vapour pressure at 50 ºC: | Non-applicable * |
| Evaporation rate at 20 ºC: | Non-applicable * |

**Product description:**

| | |
|---|---|
| Density at 20 ºC: | 1250 - 1350 kg/m³ |
| Relative density at 20 ºC: | Non-applicable * |
| Dynamic viscosity at 20 ºC: | Non-applicable * |
| Kinematic viscosity at 20 ºC: | Non-applicable * |
| Kinematic viscosity at 40 ºC: | Non-applicable * |
| Concentration: | Non-applicable * |
| pH: | 11,6 - 12,6 (at 1 %) |
| Vapour density at 20 ºC: | Non-applicable * |
| Partition coefficient n-octanol/water 20 ºC: | Non-applicable * |
| Solubility in water at 20 ºC: | Non-applicable * |
| Solubility properties: | Completely miscible |
| Decomposition temperature: | Non-applicable * |
| Melting point/freezing point: | Non-applicable * |

**Flammability:**

*Not relevant due to the nature of the product, not providing information property of its hazards.

- CONTINUED ON NEXT PAGE -

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES (continued)

| | |
|---|---|
| Flash Point: | Non Flammable (>60 ºC) |
| Flammability (solid, gas): | Non-applicable * |
| Autoignition temperature: | Non-applicable * |
| Lower flammability limit: | Non-applicable * |
| Upper flammability limit: | Non-applicable * |
| **Particle characteristics:** | |
| Median equivalent diameter: | Non-applicable |

**9.2 Other information:**

**Information with regard to physical hazard classes:**

| | |
|---|---|
| Explosive properties: | Non-applicable * |
| Oxidising properties: | Non-applicable * |
| Corrosive to metals: | H290 May be corrosive to metals. |
| Heat of combustion: | Non-applicable * |
| Aerosols-total percentage (by mass) of flammable components: | Non-applicable * |
| **Other safety characteristics:** | |
| Surface tension at 20 ºC: | Non-applicable * |
| Refraction index: | Non-applicable * |

*Not relevant due to the nature of the product, not providing information property of its hazards.

## SECTION 10: STABILITY AND REACTIVITY

**10.1 Reactivity:**

No hazardous reactions are expected because the product is stable under recommended storage conditions. See section 7.

**10.2 Chemical stability:**

Chemically stable under the indicated conditions of storage, handling and use.

**10.3 Possibility of hazardous reactions:**

Under the specified conditions, hazardous reactions that lead to excessive temperatures or pressure are not expected.

**10.4 Conditions to avoid:**

Applicable for handling and storage at room temperature:

| Shock and friction | Contact with air | Increase in temperature | Sunlight | Humidity |
|---|---|---|---|---|
| Not applicable | Not applicable | Not applicable | Not applicable | Not applicable |

Avoid strong heating.
Corrodes aluminium.
Reactions with acids.

**10.5 Incompatible materials:**

| Acids | Water | Oxidising materials | Combustible materials | Others |
|---|---|---|---|---|
| Avoid strong acids | Not applicable | Precaution | Not applicable | Not applicable |

**Other information:**

Incompatible materials: Metals.

**10.6 Hazardous decomposition products:**

Development of hydrogen when reacting with aluminium.

## SECTION 11: TOXICOLOGICAL INFORMATION

**11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008:**

- CONTINUED ON NEXT PAGE -

## SECTION 11: TOXICOLOGICAL INFORMATION (continued)

The experimental information related to the toxicological properties of the product itself is not available

**Dangerous health implications:**

In case of exposure that is repetitive, prolonged or at concentrations higher than the recommended occupational exposure limits, adverse effects on health may result, depending on the means of exposure:

A- Ingestion (acute effect):

- Acute toxicity : Based on available data, the classification criteria are not met, however, it contains substances classified as dangerous for consumption. For more information see section 3.
- Corrosivity/Irritability: Corrosive product, if it is swallowed causes burns destroying the tissues. For more information about secondary effects from skin contact see section 2.

B- Inhalation (acute effect):

- Acute toxicity : Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for inhalation. For more information see section 3.
- Corrosivity/Irritability: Prolonged inhalation of the product is corrosive to mucous membranes and the upper respiratory tract

C- Contact with the skin and the eyes (acute effect):

- Contact with the skin: Above all, skin contact may occur as fabrics of all thicknesses can be destroyed, resulting in burns. For more information on the secondary effects see section 2.
- Contact with the eyes: Produces serious eye damage after contact.

D- CMR effects (carcinogenicity, mutagenicity and toxicity to reproduction):

- Carcinogenicity: Based on available data, the classification criteria are not met. However, it contains substances classified as dangerous with carcinogenic effects. For more information see section 3.
  IARC: Non-applicable
- Mutagenicity: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.
- Reproductive toxicity: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.

E- Sensitizing effects:

- Respiratory: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous with sensitising effects. For more information see section 3.
- Skin: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.

F- Specific target organ toxicity (STOT) - single exposure:

Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.

G- Specific target organ toxicity (STOT)-repeated exposure:

- Specific target organ toxicity (STOT)-repeated exposure: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.
- Skin: Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.

H- Aspiration hazard:

Based on available data, the classification criteria are not met, as it does not contain substances classified as hazardous for this effect. For more information see section 3.

**Other information:**

Non-applicable

**Specific toxicology information on the substances:**

| Identification | Acute toxicity | | Genus |
|---|---|---|---|
| potassium hydroxide | LD50 oral | 388 mg/kg | Rat |
| CAS: 1310-58-3 | LD50 dermal | >2000 mg/kg | |
| EC: 215-181-3 | LC50 inhalation | >5 mg/L | |

- CONTINUED ON NEXT PAGE -

## SECTION 11: TOXICOLOGICAL INFORMATION (continued)

| Identification | Acute toxicity | | Genus |
|---|---|---|---|
| trisodium nitrilotriacetate | LD50 oral | 686 mg/kg | Mouse |
| CAS: 5064-31-3 | LD50 dermal | >5000 mg/kg | |
| EC: 225-768-6 | LC50 inhalation | >5 mg/L | |

**Acute Toxicity Estimate (ATE mix):**

| ATE mix | | Ingredient(s) of unknown toxicity |
|---|---|---|
| Oral | 2863,07 mg/kg (Calculation method) | 0 % |
| Dermal | >2000 mg/kg (Calculation method) | Non-applicable |
| Inhalation | >20 mg/L (4 h) (Calculation method) | Non-applicable |

**11.2 Information on other hazards:**

**Endocrine disrupting properties**

Endocrine-disrupting properties: The product fails to meet the criteria.

**Other information**

Non-applicable

## SECTION 12: ECOLOGICAL INFORMATION

The experimental information related to the eco-toxicological properties of the product itself is not available

**12.1 Toxicity:**

**Acute toxicity:**

| Identification | Concentration | | Species | Genus |
|---|---|---|---|---|
| trisodium nitrilotriacetate | LC50 | 240,4 mg/L (96 h) | Carassius auratus | Fish |
| CAS: 5064-31-3 | EC50 | 950 mg/L (24 h) | Daphnia magna | Crustacean |
| EC: 225-768-6 | EC50 | 510 mg/L (120 h) | Microcystis aeruginosa | Algae |

**Chronic toxicity:**

| Identification | Concentration | | Species | Genus |
|---|---|---|---|---|
| trisodium nitrilotriacetate | NOEC | 54 mg/L | Pimephales promelas | Fish |
| CAS: 5064-31-3 EC: 225-768-6 | NOEC | Non-applicable | | |

**12.2 Persistence and degradability:**

Behaviour in sewage treatment plants:
The product is a lye. Neutralisation is usually required before discharging a wastewater into sewage treatment plants.

**12.3 Bioaccumulative potential:**

Not available

**12.4 Mobility in soil:**

Leaking product can penetrate the soil and lead to soil and groundwater contamination.

**12.5 Results of PBT and vPvB assessment:**

Product fails to meet PBT/vPvB criteria

**12.6 Endocrine disrupting properties:**

Endocrine-disrupting properties: The product fails to meet the criteria.

**12.7 Other adverse effects:**

Not described

## SECTION 13: DISPOSAL CONSIDERATIONS

**13.1 Waste treatment methods:**

| Code | Description | Waste class (Regulation (EU) No 1357/2014) |
|---|---|---|
| 20 01 29* | detergents containing hazardous substances | Dangerous |

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## SECTION 13: DISPOSAL CONSIDERATIONS (continued)

**Type of waste (Regulation (EU) No 1357/2014):**

HP8 Corrosive, HP7 Carcinogenic

**Waste management (disposal and evaluation):**

Consult the authorized waste service manager on the assessment and disposal operations in accordance with Annex 1 and Annex 2 (Directive 2008/98/EC). As under 15 01 (2014/955/EC) of the code and in case the container has been in direct contact with the product, it will be processed the same way as the actual product. Otherwise, it will be processed as non-dangerous residue. Waste should not be disposed of to drains. See paragraph 6.2.

**Regulations related to waste management:**

In accordance with Annex II of Regulation (EC) No 1907/2006 (REACH) the community or state provisions related to waste management are stated

Community legislation: Directive 2008/98/EC, 2014/955/EU, Regulation (EU) No 1357/2014

## SECTION 14: TRANSPORT INFORMATION

**Transport of dangerous goods by land:**

With regard to ADR 2021 and RID 2021:


| | | |
|---|---|---|
| **14.1** | **UN number or ID number:** | UN1814 |
| **14.2** | **UN proper shipping name:** | POTASSIUM HYDROXIDE SOLUTION |
| **14.3** | **Transport hazard class(es):** | 8 |
| | Labels: | 8 |
| **14.4** | **Packing group:** | II |
| **14.5** | **Environmental hazards:** | No |
| **14.6** | **Special precautions for user** | |
| | Special regulations: | Non-applicable |
| | Tunnel restriction code: | E |
| | Physico-Chemical properties: | see section 9 |
| | Limited quantities: | 1 L |
| **14.7** | **Maritime transport in bulk according to IMO instruments:** | Non-applicable |

**Transport of dangerous goods by sea:**

With regard to IMDG 40-20:


| | | |
|---|---|---|
| **14.1** | **UN number or ID number:** | UN1814 |
| **14.2** | **UN proper shipping name:** | POTASSIUM HYDROXIDE SOLUTION |
| **14.3** | **Transport hazard class(es):** | 8 |
| | Labels: | 8 |
| **14.4** | **Packing group:** | II |
| **14.5** | **Marine pollutant:** | No |
| **14.6** | **Special precautions for user** | |
| | Special regulations: | Non-applicable |
| | EmS Codes: | F-A, S-B |
| | Physico-Chemical properties: | see section 9 |
| | Limited quantities: | 1 L |
| | Segregation group: | SGG18 |
| **14.7** | **Maritime transport in bulk according to IMO instruments:** | Non-applicable |

**Transport of dangerous goods by air:**

With regard to IATA/ICAO 2023:

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## SECTION 14: TRANSPORT INFORMATION (continued)


| | | |
|---|---|---|
| **14.1** | **UN number or ID number:** | UN1814 |
| **14.2** | **UN proper shipping name:** | POTASSIUM HYDROXIDE SOLUTION |
| **14.3** | **Transport hazard class(es):** | 8 |
| | Labels: | 8 |
| **14.4** | **Packing group:** | II |
| **14.5** | **Environmental hazards:** | No |
| **14.6** | **Special precautions for user** | |
| | Physico-Chemical properties: | see section 9 |
| **14.7** | **Maritime transport in bulk according to IMO instruments:** | Non-applicable |

## SECTION 15: REGULATORY INFORMATION

**15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture:**

Candidate substances for authorisation under the Regulation (EC) No 1907/2006 (REACH): Non-applicable

Substances included in Annex XIV of REACH ("Authorisation List") and sunset date: Non-applicable

Regulation (EC) No 1005/2009, about substances that deplete the ozone layer: Non-applicable

Article 95, REGULATION (EU) No 528/2012: Non-applicable

REGULATION (EU) No 649/2012, in relation to the import and export of hazardous chemical products: Non-applicable

**Regulation (EC) No 648/2004 on detergents:**

In accordance with this regulation the product complies with the following:

**Labelling for contents:**

| Component | Concentration interval |
|---|---|
| Phosphonates | % (w/w) < 5 |
| NTA (nitrilotriacetic acid) and salts thereof | % (w/w) < 5 |

**Seveso III:**

Non-applicable

**Limitations to commercialisation and the use of certain dangerous substances and mixtures (Annex XVII REACH, etc ….):**

Shall not be used in:
—ornamental articles intended to produce light or colour effects by means of different phases, for example in ornamental lamps and ashtrays,
—tricks and jokes,
—games for one or more participants, or any article intended to be used as such, even with ornamental aspects.

**Specific provisions in terms of protecting people or the environment:**

It is recommended to use the information included in this safety data sheet as a basis for conducting workplace-specific risk assessments in order to establish the necessary risk prevention measures for the handling, use, storage and disposal of this product.

**Other legislation:**

The product could be affected by sectorial legislation

- Regulation (EC) No 1223/2009 of the European Parliament and of the Council of 30 November 2009 on cosmetic products
- Regulation (EC) No 648/2004 of the European Parliament and of the Council of 31 March 2004 on detergents
- Commission Regulation (EC) No 907/2006 of 20 June 2006 amending Regulation (EC) No 648/2004 of the European Parliament and of the Council on detergents, in order to adapt Annexes III and VII
- Commission Regulation (EC) No 551/2009 of 25 June 2009 amending Regulation (EC) No 648/2004 of the European Parliament and of the Council on detergents, in order to adapt Annexes V and VI thereto (surfactant derogation)

**15.2 Chemical safety assessment:**

The supplier has not carried out evaluation of chemical safety.

- CONTINUED ON NEXT PAGE -

## SECTION 16: OTHER INFORMATION

**Legislation related to safety data sheets:**

The SDS shall be supplied in an official language of the country where the product is placed on the market. This safety data sheet has been designed in accordance with ANNEX II-Guide to the compilation of safety data sheets of Regulation (EC) No 1907/2006 (COMMISSION REGULATION (EU) 2020/878).

**Modifications related to the previous Safety Data Sheet which concerns the ways of managing risks.:**

Non-applicable

**Texts of the legislative phrases mentioned in section 2:**

H290: May be corrosive to metals.
H314: Causes severe skin burns and eye damage.
H318: Causes serious eye damage.

**Texts of the legislative phrases mentioned in section 3:**

The phrases indicated do not refer to the product itself; they are present merely for informative purposes and refer to the individual components which appear in section 3

**CLP Regulation (EC) No 1272/2008:**

Acute Tox. 4: H302 - Harmful if swallowed.
Carc. 2: H351 - Suspected of causing cancer.
Eye Irrit. 2: H319 - Causes serious eye irritation.
Met. Corr. 1: H290 - May be corrosive to metals.
Skin Corr. 1A: H314 - Causes severe skin burns and eye damage.

**Classification procedure:**

Skin Corr. 1: Calculation method
Eye Dam. 1: Calculation method

**Advice related to training:**

Training is recommended in order to prevent industrial risks for staff using this product and to facilitate their comprehension and interpretation of this safety data sheet, as well as the label on the product.

**Principal bibliographical sources:**

http://echa.europa.eu
http://eur-lex.europa.eu

**Abbreviations and acronyms:**

ADR: European agreement concerning the international carriage of dangerous goods by road
IMDG: International maritime dangerous goods code
IATA: International Air Transport Association
ICAO: International Civil Aviation Organisation
COD: Chemical Oxygen Demand
BOD5: 5day biochemical oxygen demand
BCF: Bioconcentration factor
LD50: Lethal Dose 50
LC50: Lethal Concentration 50
EC50: Effective concentration 50
LogPOW: Octanolwater partition coefficient
Koc: Partition coefficient of organic carbon
UFI: unique formula identifier
IARC: International Agency for Research on Cancer



- END OF SAFETY DATA SHEET -

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BEST TECHNOLOGY SOLUTION with schekolin
Hochwertige Lacke und Farben
High quality lacquers and paints

Informazioni Tecniche
Technical Information Sheet
Technische Informationen
Informations techniques

**Rev. 01 – 30.05.2022**

# 422 9 918 57512 SCHEKOSOL CLEAR BC

**Suggested use :** Colourless Basecoat for exterior of monobloc Aerosol cans
**Type :** Polyester
**Solvents :** Aromatic hydrocarbons, Alcohols, Glycolethers, Glycolesters

## PHYSICAL CHARACTERISTICS

| | | | |
|---|---|---|---|
| **Delivery Viscosity :** | (DIN 6 cup at 20 °C) | 50 ± 5 | sec. |
| **Application Viscosity :** | (DIN 6 cup at 20 °C) | 45 – 90 | sec. |
| **Density :** | (Pycnometer at 20 °C) | 1010 – 1060 | g/l |
| **Solid contents :** | (30 min. at 180 °C) | 51 ± 2 | % |
| **Shelf life :** | (At about 25 °C) | 6 months | |
| | *-store in cool and dry place in original packaging-* | | |

## RECOMMENDED APPLICATION PROCEDURE

| | | |
|---|---|---|
| **Apply by :** | Roller coater | |
| **Apply over :** | Monobloc aerosol cans | |
| **Preliminary treatment :** | Usual pre-treatment | |
| **Dilution :** | Usually not needed, 1 – 2 % if necessary | |
| **Reducer :** | 506 9 003 | |
| **Wet film weight :** | 25 – 31 $g/m^2$ | |
| **Dry film weight :** | 10 – 12 µm | |
| **Curing conditions :** | Pre-drying | 3 – 7 min. at 100 – 120 °C |
| | Intermediate drying | 5 – 7 min. at 150 – 160 °C |
| | Final-drying | 6 – 10 min. at 160 – 180 °C |

**REMARKS :**

- Good elasticity
- Steam bath resistant
- Excellent mechanical properties
- Very good levelling
- Good sweating effect
- Recommended procedure: 1$^{st}$ Schekosol BC 4229918, 2$^{nd}$ Printing, 3$^{rd}$ Suitable overprint varnish, 4$^{th}$ Final drying

**DEVELOPMENT AND QUALITY CONTROL TEST ARE PERFORMED ON THE FOLLOWING SUBSTRATES:**
**ETP E 2.0/2.0 or E 2.8/2.8 upon availability, stone finished, 310 or 311 passivated; ECCS with min. 50 mg/sqm of Chrome; ALUMINIUM 8011 alloy, H14 temper**

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*Rev. 01 – 30.05.2022*

# 422 9 903 57705 SCHEKOSOL OPV SILKMATT

| | |
|---|---|
| **Suggested use :** | Silk matt Overprint varnish for exterior of Aerosol cans made of aluminium |
| **Type :** | Polyester |
| **Solvents :** | Aromatic hydrocarbons, Alcohols, Glycolesters |

## PHYSICAL CHARACTERISTICS

| | | | |
|---|---|---|---|
| **Delivery Viscosity :** | (DIN 4 cup at 20 °C) | 85 ± 5 | sec. |
| **Application Viscosity :** | (DIN 4 cup at 20 °C) | 80 – 100 | sec. |
| **Density :** | (Pycnometer at 20 °C) | 1050 – 1090 | g/l |
| **Solid contents :** | (30 min. at 170 °C) | 47 ± 2 | % |
| **Shelf life :** | (At about 25 °C) | 6 months | |
| | *-store in cool and dry place in original packaging-* | | |

## RECOMMENDED APPLICATION PROCEDURE

| | |
|---|---|
| **Apply by :** | Roller coater |
| **Apply over :** | Pre-coated and printed Aerosol can |
| **Preliminary treatment :** | Usual pre-treatment |
| **Dilution :** | Usually not needed, 1 – 2 % if necessary |
| **Reducer :** | 506 9 003 |
| **Wet film weight :** | 11 – 17 $g/m^2$ |
| **Dry film weight :** | 4 – 6 µm |
| **Curing conditions :** | 8 - 10 min. PMT at 160-170 °C |

## REMARKS :

- Good hardness
- Resistant to chemicals
- Good abrasion resistance
- Perfectly contractible and borderfast
- Recommended procedure: 1st Suitable basecoat, 2nd Printing, 3rd Drying, 4th Schekosol OPV 4229903, 5th Final drying

**DEVELOPMENT AND QUALITY CONTROL TEST ARE PERFORMED ON THE FOLLOWING SUBSTRATES:**
**ETP E 2.0/2.0 or E 2.8/2.8 upon availability, stone finished, 310 or 311 passivated; ECCS with min. 50 mg/sqm of Chrome; ALUMINIUM 8011 alloy, H14 temper**

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BEST TECHNOLOGY SOLUTION with schekolin — Hochwertige Lacke und Farben — High quality lacquers and paints

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*Rev. 01 – 30.05.2022*

# 422 9 900 — 57704 SCHEKOSOL OPV GLOSSY

| | |
|---|---|
| **Suggested use :** | Colourless, Glossy overprint varnish for aerosol cans |
| **Type :** | Polyester |
| **Solvents :** | Aromatic Hydrocarbons, Alcohols, Glycolesters |

## PHYSICAL CHARACTERISTICS

| | | | |
|---|---|---|---|
| **Delivery Viscosity :** | (DIN 4 cup at 20 °C) | 90 ± 5 | s. |
| **Application Viscosity :** | (DIN 4 cup at 20 °C) | 90 ± 5 | s. |
| **Density :** | (Pycnometer at 20 °C) | 1050 – 1090 | g/L |
| **Solid contents :** | (30 min. at 180 °C) | 43 ± 2 | % |
| **Shelf life :** | (At about 25 °C) | 6 months | |
| | *-store in cool and dry place in original packaging-* | | |

## RECOMMENDED APPLICATION PROCEDURE

| | |
|---|---|
| **Apply by :** | Roller coating |
| **Apply over :** | Precoated and printed aluminium monobloc aerosol cans |
| **Preliminary treatment :** | Usual pre-treatment |
| **Dilution :** | Usually not needed, 1 – 2 % if necessary |
| **Reducer :** | 506 9 003 |
| **Wet film weight :** | 13 – 19 g/m$^2$ |
| **Dry film weight :** | 4 – 6 µm |
| **Curing conditions :** | 8 – 10 min. at 160 – 170 °C |

## REMARKS :

- Excellent flexibility
- Resistant to chemicals
- Good abrasion resistance
- Resistant to water bath
- Recommended procedure: 1$^{st}$ Schekosol basecoat for aerosol cans, 2$^{nd}$ Printing, 3$^{rd}$ overprint varnish 4229900, 4$^{th}$ Final drying

**DEVELOPMENT AND QUALITY CONTROL TEST ARE PERFORMED ON THE FOLLOWING SUBSTRATES:**
**ETP E 2.0/2.0 or E 2.8/2.8 upon availability, stone finished, 310 or 311 passivated; ECCS with min. 50 mg/sqm of Chrome; ALUMINIUM 8011 alloy, H14 temper**